



PIYUSH MAJI
Master of Science in Physics
IIT Hyderabad, Telangana, India

+91 9330966439
ph25mscst11019@iith.ac.in
LinkedIn GitHub
piyush314159.github.io

EDUCATION

Indian Institute of Technology Hyderabad

Expected July 2027

M.Sc. in Physics CGPA: 7.5/10

Hooghly Mohsin College (1836), West Bengal

August 2024

B.Sc. in Physics CGPA: 7.28/10

RESEARCH

GNN-Based Full Event Interpretation (GraFEI) — Belle II

Ongoing

Prof. Sourav Sandilya, Department of Physics, IIT Hyderabad

- Studying ML-driven inclusive event reconstruction for the Belle II experiment at KEK using the Graph-based Full Event Interpretation (GraFEI) framework, which applies deep GNNs to reconstruct full decay topologies from final state particles alone, without prior assumptions on the decay chain structure.
- Building detector-level graph representations of particle decay chains using Python and PyTorch Geometric; modelling each reconstructed particle as a graph node with kinematic features (momentum, DOCA, $\cos\theta$) and learning the Lowest Common Ancestor Stage (LCAS) matrix encoding the decay hierarchy ($\Upsilon(4S) \rightarrow B \rightarrow D^* \rightarrow \dots$).
- Training the GraFEIModel (stacked MetaLayers with EdgeLayer, NodeLayer, GlobalLayer) to jointly predict particle mass hypotheses and the LCAS matrix via a combined cross-entropy loss $\mathcal{L} = \mathcal{L}_{LCA} + \alpha \mathcal{L}_{Masses}$; evaluating reconstruction quality using PerfectLCA, PerfectMasses, and PerfectEvent metrics within the basf2 analysis framework.

PROJECTS

Spam Classifier from Scratch

Completed

Python, NumPy, Machine Learning [GitHub](#)

- Built Multinomial Naive Bayes (MLE + Laplace smoothing) and Logistic Regression (gradient descent) entirely from scratch in NumPy on the UCI SMS Spam Collection (5,574 messages); implemented both TF-IDF and Bag-of-Words feature pipelines without sklearn, with vocabulary built strictly on the train split to prevent data leakage.
- Implemented binary cross-entropy loss and hand-coded gradient descent, visualising real-time loss convergence; evaluated both models with precision, recall, F1, and ROC/AUC computed via the trapezoidal rule; analysed threshold sensitivity (0.3–0.7) to explore the precision-recall tradeoff on an 86.6% class-imbalanced dataset.

Friday — Voice Assistant

Completed

Python, OpenAI API, SpeechRecognition, pyttsx3 [GitHub](#)

- Built a wake-word activated voice assistant with modular action routing for browser control, YouTube playback, live news fetching (NewsAPI), Wikipedia search, and real-time Q&A via OpenAI API fallback.
- Implemented speech-to-text (Google Speech Recognition) and TTS (pyttsx3) with thread-safe audio playback; designed an intent-matching command router with extensible plugin-style action modules.

COVID-19 Epidemic Modeling — SIR / SEIR / SUTRA

Completed

Prof. Anupam Gupta, IIT Hyderabad [GitHub](#)

- Implemented SIR, SEIR, and SUTRA compartmental models on US COVID-19 data (Feb 2020–Mar 2021); used NNLS regression for per-phase parameter estimation, achieving MAPE $\sim 4\%$ on individual phases; ensembled all three models via linear regression for improved generalisation.

RELEVANT COURSES

Machine Learning for Physical Sciences • Machine Learning for High Energy Physics • Statistical Mechanics • Linear Algebra • Computational Physics • Data Science Analysis

SKILLS

Languages & Tools: Python, Git, NumPy, Pandas, PyTorch, PyTorch Geometric, basf2

Domain: Machine Learning, Graph Neural Networks, Scientific Computing, Data Science, Data Structures & Algorithms

Coding Practice: LeetCode, GeeksforGeeks, Striver A2Z, Codeforces

LEADERSHIP & EXTRACURRICULAR

Placement Coordinator

May 2026 – Present

Office of Career Services, IIT Hyderabad

Managing end-to-end campus recruitment drives, liaising with recruiting companies, and overseeing placement operations for the student body.

Inbound & Logistics Management

Sep 2025 – Present

International Relations Cell, IIT Hyderabad

Coordinating orientations, cultural programs, and academic tours for international exchange students from Japan and the United States; managing logistics for visiting international guests.

Design Core

May 2026 – Present

Extra Mural Lectures (EML), IIT Hyderabad

Leading visual design and multimedia outreach for EML; creating promotional content, event branding, and design assets for club initiatives and student entrepreneurship programs.